

JavaScript Function Challenges

Challenge 1: Function Declaration

Write a function `sayHello` that prints `"Hello, Developer!"`.

```
```js
// Your code here

sayHello(); // Output: Hello, Developer!
```
```

Challenge 2: Function Expression

Create a function expression that returns the square of a number.

```
```js
// Your code here

console.log(square(4)); // Output: 16
```
```

Challenge 3: Arrow Function

Convert the following function to an arrow function.

```
```js
function multiply(a, b) {
 return a * b;
}

// Convert to arrow and call with (3, 4)
```
```

Challenge 4: Named Function Expression

Create a named function expression called `factorial` that returns the factorial of a number.

```
```js
// Your code here

console.log(fact(5)); // Output: 120
```
```

Challenge 5: Anonymous Function

Pass an anonymous function as a callback to `setTimeout` that logs `"Timer Done"` after 1 second.

```
```js
```

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```
// Your code here
```
```

Challenge 6: IIFE

Create an IIFE that logs ``"App started"`` as soon as the script runs.

```
```js
// Your code here
```
```

Challenge 7: Constructor Function

Create a constructor function ``Car`` with ``brand`` and ``model``, and a method to display them.

```
```js
// Your code here

const myCar = new Car("Toyota", "Corolla");
myCar.display(); // Output: Toyota Corolla
```
```

Challenge 8: Generator Function

Write a generator that yields the numbers 1 to 3.

```
```js
// Your code here

const gen = numberGen();
console.log(gen.next().value); // 1
console.log(gen.next().value); // 2
console.log(gen.next().value); // 3
```
```

Challenge 9: Async Function

Write an async function ``getMessage()`` that returns ``"Hello Async"`` using a Promise.

```
```js
// Your code here

getMessage().then(console.log); // Output: Hello Async
```
```

Challenge 10: Callback Function

Create a function ``process`` that takes a number and a callback. The callback should

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double the number.

```
```js
// Your code here

process(5, function (num) {
 console.log(num); // Output: 10
});
```
```

Challenge 11: Higher-Order Function

Write a function `makeLouder` that takes a string function and returns a new function that converts its output to uppercase.

```
```js
function sayHi() {
 return "hi";
}

// Your code here

const loudHi = makeLouder(sayHi);
console.log(loudHi()); // Output: HI
```
```

Challenge 12: Object Method

Add a method `greet` to the object that uses `this.name` to greet.

```
```js
const person = {
 name: "Krian",
 // Your code here
};

person.greet(); // Output: Hello, Krian
```
```

Challenge 13: Class Method

Define a class `User` with a constructor and a method `sayName` that logs the user's name.

```
```js
// Your code here

const u = new User("Kiran");
u.sayName(); // Output: My name is Kiran
```
```